Presentation 2

| 2025.10.23

LEGAL AID CHICAGO

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Objective 1.1

Methods

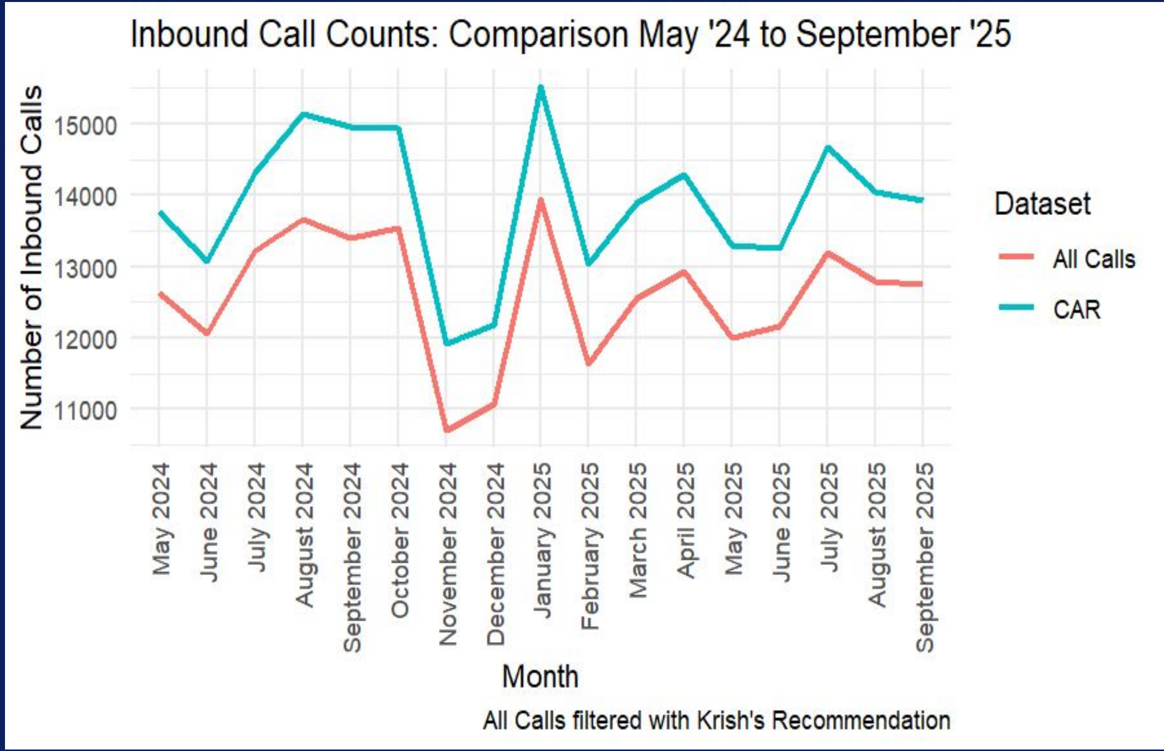
CAR

- Keep all distinct Contact Session IDs

All Calls

- Implemented both Krish and Edwin Rong's strategies for identifying inbound calls (see next slide)
- Krish: duration > 0, terminating direction, "CallTower" PSTN Vendor, 6 phone lines, unique correlation IDs
- Edwin: inbound trunk is NA, outbound trunk starting with "wcc", SIP INBOUND call type, and WXCC client type

Correct the consistency-check plots for the number of inbound calls



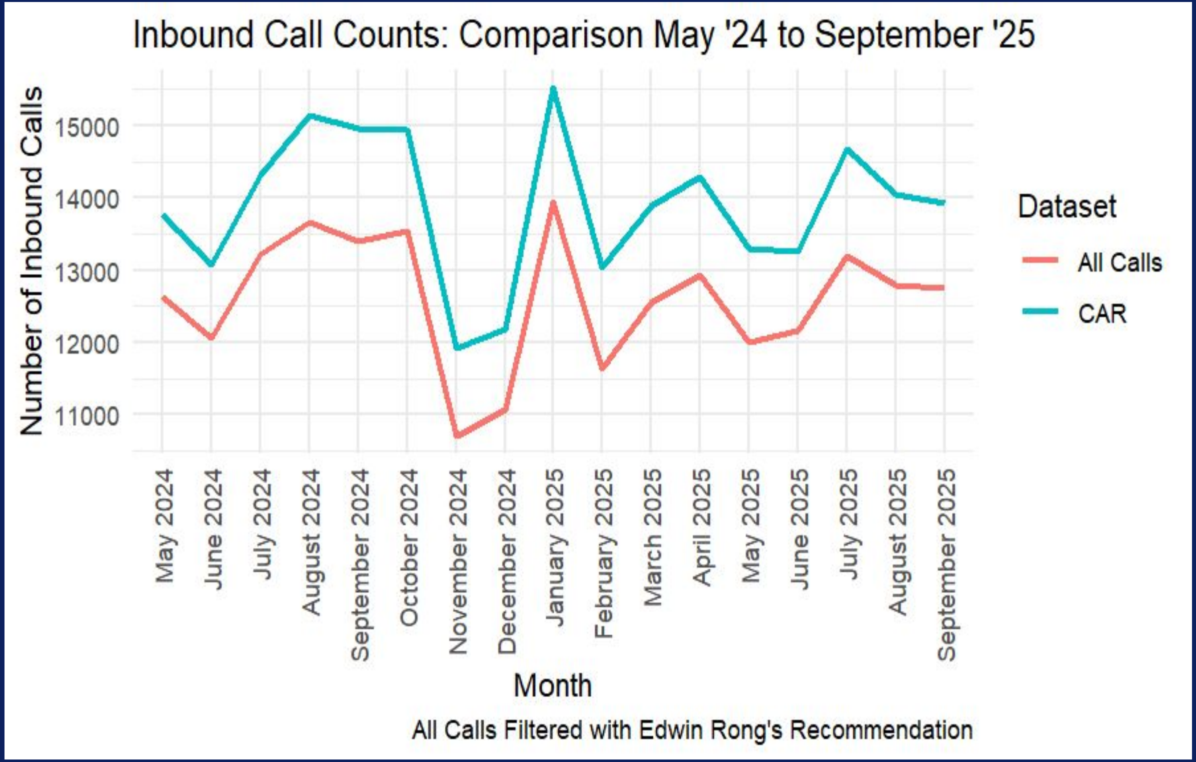
Objective 1.1

Takeaways

- Both methods for identifying inbound calls in All Calls leave us with a similar difference from CAR.
- Despite this disparity, it is encouraging that both datasets demonstrate a similar trend over the months.
- Still need to identify potential causes for the difference – especially in months such as August 2024 where the trend is visibly steeper in CAR.

Both filtering strategies for All Calls produce similar results, so seems like the issue lies in how we're filtering CAR.

Correct the consistency-check plots for the number of inbound calls



Objective 1.2

Make a consistency check for inbound call duration between CAR and All Calls datasets after March 15, 2025

Method: extract the hour of day from call timestamps and grouped median call durations by hour to visualize how call length patterns varied throughout the day

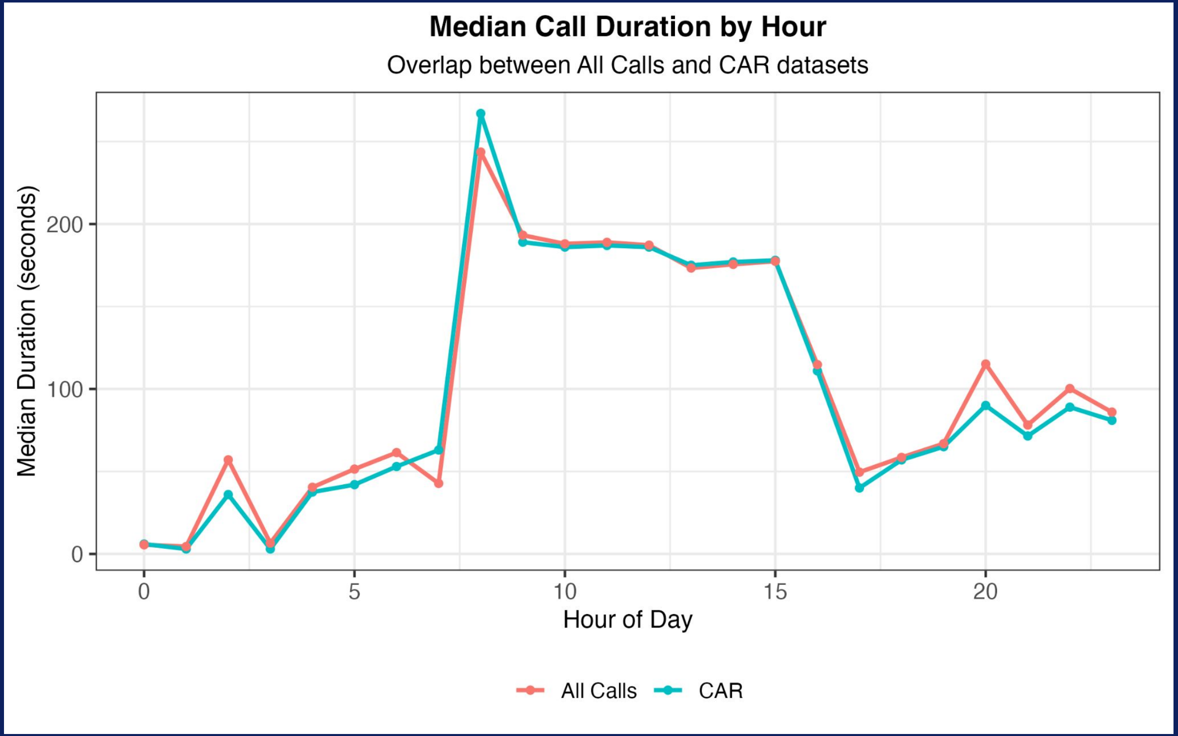
<table>
<tr><th></th><th>CAR</th><th>All Calls</th></tr>
<tr><td>Unique record key</td><td>• contact_session_id</td><td>• correlation_id + called_number</td></tr>
<tr><td>Duration calculation</td><td>Time difference between the max and min timestamps within each session</td><td>Time differences between release_time & answer_time and release_time & start_time</td></tr>
<tr><td>Hour assignment</td><td>• start_time before 17:00
• end_time after 17:00</td><td>• report_time</td></tr>
<tr><td>Physical code</td><td><code>car_durations <- car |>
  summarize(
    .by = contact_session_id,
    start_time = min(activity_start_timestamp, na.rm = TRUE),
    end_time   = max(activity_start_timestamp, na.rm = TRUE),
    duration   = as.numeric(end_time - start_time, units = "secs")
) |>
  mutate(
    # derive hour from start_time and end_time
    start_hour = hour(start_time),
    end_hour   = hour(end_time),
    # use start hour before 17:00, otherwise use end hour (evening calls often span hours)
    hour = if_else(start_hour < 17, start_hour, end_hour)
) |></code></td><td><code>distinct(correlation_id, called_number, .keep_all = TRUE) |>
mutate(
  duration_release_time_answer_time = as.numeric(release_time - answer_time),
  duration_release_time_start_time = as.numeric(release_time - start_time),
  hour = hour(report_time)
) |></code></td></tr>
</table>

Objective 1.2

Make a consistency check for inbound call duration between CAR and All Calls datasets after March 15, 2025

Takeaways

- CAR and All Calls datasets show **consistent hourly patterns**, indicating reliable overlap and alignment between the two sources
- Median call durations peak around 8 AM, the start of operational hours, gradually decline toward the end of the workday
- CAR: the hour variable was assigned using the start hour for calls before 17:00 and the end hour for calls afterward, ensuring that **evening or extended calls** were categorized more accurately within the correct operational period
- All Calls: answer_time and start_time don't make a difference



Objective 2

Identify a set of scenarios where the call ending is positive, negative, or neutral to help visualize the activity when a customer leaves

Positive Outcomes

Related to Termination Reason:

- Agent ends call assuming issue has been resolved
  - 'Agent Left'
  - 'AGENT_ENDS'

Additional Reasons:

- Customer leaves call once their issue has been addressed
  - Will have to look at subsection of Customer Left that makes it to an agent
- Agent or customer leaves at the end of successful courtesy callback
  - Difficult to tell from termination reason alone as there may be multiple terminations

Neutral Outcomes

Related to Termination Reason:

- Customer is unreachable
  - 'NO_ANSWER_FROM_CUSTOMER'
  - 'NO_ANSWER_CUSTOMER'
  - 'CUSTOMER_BUSY'
  - 'MAX_CALLBACK_RETRY_LIMIT_REACHED'
  - 'Participant Invite timer expired'
  - 'CUSTOMER_UNAVAILABLE'
  - 'USER_DECLINED'
  - 'USER_UNAVAILABLE'
  - 'USER_DECLINED'
  - 'USER_BUSY'
  - 'NO_ANSWER_USER'
  - 'QUEUE_TIMEOUT'

Additional Reasons:

- Customer calls while LAC is closed
- LAC does not handle customer's legal issue
- Queue Timeout with courtesy call back option chosen

Negative Outcomes

Related to Termination Reason:

- Agent is unable to receive call
  - 'AGENT_UNAVAILABLE'
  - 'AGENT_BUSY'
  - 'NO_ANSWER_FROM_AGENT'
- System issue or failure
  - 'OUTDIAL_FAILED'
  - 'MEDIA_MANAGER_INTERNAL_ERROR'
  - 'CHANNEL_FAILURE'

Additional Reasons:

- Customer is unable to effectively navigate to correct menu or queue for their legal issue
- Customer enters a queue, but wait time is too long and they leave

Objective 2

Considering an abandoned call to be a negative outcome with ‘Customer Left’ as the termination reason

Potential Cases for Abandoned Calls

- Not listening to all 300 seconds of hold music
- Last row for call has an activity name
- Customer has an extended journey through many menus, but never enters a queue, listens to a voicemail, or talks to an agent
- Call is very short & customer never enters a queue → possibility to be neutral or negative
  - Cases:
    - Customer changes their mind → neutral
    - Customer is overwhelmed/confused → negative
    - Customer should not be able to get to a queue (e.g. closed) → neutral
- Customer enters queue but wait time is too long and either:
  - Queue closes, or
  - Customer leaves due to wait time

Objective 2

	index	Contact Session ID	Activity Start Timestamp	Step Category	Activity
70933	143963	0d5dec0a-ec28-4bb0-a728-5f848b543d8d	2025/03/23 05:25:35 AM	Queue Name	Family
70934	143964	0d5dec0a-ec28-4bb0-a728-5f848b543d8d	2025/03/23 05:25:35 AM	Queue Name	Family
77159	191941	0d5dec0a-ec28-4bb0-a728-5f848b543d8d	2025/03/23 05:25:35 AM	Termination Reason	Customer Left
77160	191942	0d5dec0a-ec28-4bb0-a728-5f848b543d8d	2025/03/23 05:25:35 AM	Termination Reason	Queue Timeout
12	31	0d5dec0a-ec28-4bb0-a728-5f848b543d8d	2025/03/23 05:25:45 AM	EP Name	Courtesy Callback Telephony EP
13	34	0d5dec0a-ec28-4bb0-a728-5f848b543d8d	2025/03/23 05:25:45 AM	EP Name	Courtesy Callback Telephony EP
49394	95987	0d5dec0a-ec28-4bb0-a728-5f848b543d8d	2025/03/23 05:25:45 AM	Activity Name	CallbackRetry
70935	143966	0d5dec0a-ec28-4bb0-a728-5f848b543d8d	2025/03/23 05:25:45 AM	Queue Name	Family
70936	143967	0d5dec0a-ec28-4bb0-a728-5f848b543d8d	2025/03/23 05:25:45 AM	Queue Name	Family
70937	143968	0d5dec0a-ec28-4bb0-a728-5f848b543d8d	2025/03/23 05:25:45 AM	Queue Name	Family
77161	191945	0d5dec0a-ec28-4bb0-a728-5f848b543d8d	2025/03/23 05:25:45 AM	Termination Reason	Customer Left
77162	191946	0d5dec0a-ec28-4bb0-a728-5f848b543d8d	2025/03/23 05:25:45 AM	Termination Reason	Customer Left
70979	144262	0d5dec0a-ec28-4bb0-a728-5f848b543d8d	2025/03/24 02:25:45 AM	Queue Name	Family
70980	144263	0d5dec0a-ec28-4bb0-a728-5f848b543d8d	2025/03/24 02:25:45 AM	Queue Name	Family
77218	192240	0d5dec0a-ec28-4bb0-a728-5f848b543d8d	2025/03/24 02:25:45 AM	Termination Reason	Queue Timeout
77219	192241	0d5dec0a-ec28-4bb0-a728-5f848b543d8d	2025/03/24 02:25:45 AM	Termination Reason	Customer Left
231	330	0d5dec0a-ec28-4bb0-a728-5f848b543d8d	2025/03/24 02:25:55 AM	EP Name	Courtesy Callback Telephony EP
232	333	0d5dec0a-ec28-4bb0-a728-5f848b543d8d	2025/03/24 02:25:55 AM	EP Name	Courtesy Callback Telephony EP
49486	96286	0d5dec0a-ec28-4bb0-a728-5f848b543d8d	2025/03/24 02:25:55 AM	Activity Name	CallbackRetry
70981	144265	0d5dec0a-ec28-4bb0-a728-5f848b543d8d	2025/03/24 02:25:55 AM	Queue Name	Family
70982	144266	0d5dec0a-ec28-4bb0-a728-5f848b543d8d	2025/03/24 02:25:55 AM	Queue Name	Family
70983	144267	0d5dec0a-ec28-4bb0-a728-5f848b543d8d	2025/03/24 02:25:55 AM	Queue Name	Family
77220	192244	0d5dec0a-ec28-4bb0-a728-5f848b543d8d	2025/03/24 02:25:55 AM	Termination Reason	Customer Left
77221	192245	0d5dec0a-ec28-4bb0-a728-5f848b543d8d	2025/03/24 02:25:55 AM	Termination Reason	Customer Left
1029	1410	0d5dec0a-ec28-4bb0-a728-5f848b543d8d	2025/03/24 08:33:14 AM	EP Name	Courtesy Callback Telephony EP
71088	145344	0d5dec0a-ec28-4bb0-a728-5f848b543d8d	2025/03/24 08:33:14 AM	Queue Name	Family
77285	193322	0d5dec0a-ec28-4bb0-a728-5f848b543d8d	2025/03/24 08:33:14 AM	Termination Reason	Customer Left
1030	1411	0d5dec0a-ec28-4bb0-a728-5f848b543d8d	2025/03/24 08:33:15 AM	EP Name	Courtesy Callback Telephony EP
50033	97367	0d5dec0a-ec28-4bb0-a728-5f848b543d8d	2025/03/24 08:33:15 AM	Activity Name	LegalServerScreenPop
1051	1439	0d5dec0a-ec28-4bb0-a728-5f848b543d8d	2025/03/24 08:33:55 AM	EP Name	Courtesy Callback Telephony EP
71089	145373	0d5dec0a-ec28-4bb0-a728-5f848b543d8d	2025/03/24 08:33:55 AM	Queue Name	Family
77286	193351	0d5dec0a-ec28-4bb0-a728-5f848b543d8d	2025/03/24 08:33:55 AM	Termination Reason	Customer Left
1061	1453	0d5dec0a-ec28-4bb0-a728-5f848b543d8d	2025/03/24 08:34:21 AM	EP Name	Courtesy Callback Telephony EP
71090	145387	0d5dec0a-ec28-4bb0-a728-5f848b543d8d	2025/03/24 08:34:21 AM	Queue Name	Family
71091	145388	0d5dec0a-ec28-4bb0-a728-5f848b543d8d	2025/03/24 08:34:21 AM	Queue Name	Family
77288	193365	0d5dec0a-ec28-4bb0-a728-5f848b543d8d	2025/03/24 08:34:21 AM	Termination Reason	Agent Left
71100	145435	0d5dec0a-ec28-4bb0-a728-5f848b543d8d	2025/03/24 08:36:21 AM	Queue Name	Family

Courtesy Callbacks

- For many calls, a person will wait in a queue and opt in to receiving a courtesy callback
  - This would be considered neutral
- Since courtesy callbacks can be traced with the same Contact ID, we can see how many calls it took for a different outcome to occur
- For a callback to be positive, customer must:
  - Speak to an agent
  - Have either customer left or agent left as termination reason
- For a call back to be negative, customer must:
  - Have “MAX_CALLBACK_RETRY_LIMIT_REACHED” as termination reason
  - USER_DECLINED, USER_UNAVAILABLE

There are 118 different clients getting a callback

Code for identifying each outcome can be found in appendix

Objective 2

System Disconnected Contact

- This Termination Reason is assigned to calls that occur during the closed holiday hours
- Thus we can assume this to be negative

Contact Session ID	Activity Start Timestamp	Step Category	Activity
fdc56f2f-75b9-4642-8fcb-f05b7ba668b8	2025/03/26 05:37:10 PM	EP Name	Main Number Telephony EP
fdc56f2f-75b9-4642-8fcb-f05b7ba668b8	2025/03/26 05:37:10 PM	EP Name	Main Number Telephony EP
fdc56f2f-75b9-4642-8fcb-f05b7ba668b8	2025/03/26 05:37:10 PM	EP Name	Main Number Telephony EP
fdc56f2f-75b9-4642-8fcb-f05b7ba668b8	2025/03/26 05:37:10 PM	Flow Name	LACMain
fdc56f2f-75b9-4642-8fcb-f05b7ba668b8	2025/03/26 05:37:10 PM	Flow Name	LACMain
fdc56f2f-75b9-4642-8fcb-f05b7ba668b8	2025/03/26 05:37:10 PM	Activity Name	LanguageSelectionMenu
fdc56f2f-75b9-4642-8fcb-f05b7ba668b8	2025/03/26 05:37:28 PM	EP Name	Main Number Telephony EP
fdc56f2f-75b9-4642-8fcb-f05b7ba668b8	2025/03/26 05:37:28 PM	Activity Name	LanguageSelectionMenu
fdc56f2f-75b9-4642-8fcb-f05b7ba668b8	2025/03/26 05:37:41 PM	EP Name	Closed Hours-Holidays Menu Telephony EP
fdc56f2f-75b9-4642-8fcb-f05b7ba668b8	2025/03/26 05:37:41 PM	Flow Name	ClosedHoursHolidaysMenu
fdc56f2f-75b9-4642-8fcb-f05b7ba668b8	2025/03/26 05:37:41 PM	Activity Name	ClosedMenu
fdc56f2f-75b9-4642-8fcb-f05b7ba668b8	2025/03/26 05:38:35 PM	EP Name	Closed Hours-Holidays Menu Telephony EP
fdc56f2f-75b9-4642-8fcb-f05b7ba668b8	2025/03/26 05:38:35 PM	EP Name	Closed Hours-Holidays Menu Telephony EP
fdc56f2f-75b9-4642-8fcb-f05b7ba668b8	2025/03/26 05:38:35 PM	EP Name	Closed Hours-Holidays Menu Telephony EP
fdc56f2f-75b9-4642-8fcb-f05b7ba668b8	2025/03/26 05:38:35 PM	Activity Name	DisconnectContact1
fdc56f2f-75b9-4642-8fcb-f05b7ba668b8	2025/03/26 05:38:35 PM	Termination Reason	System disconnected the contact
fdc56f2f-75b9-4642-8fcb-f05b7ba668b8	2025/03/26 05:38:35 PM	Termination Reason	System disconnected the contact

There are 3685 calls that are not served due to closed hours

Code for identifying each outcome can be found in appendix

Objective 2

Calls abandoned during hold music :

24	37	25390a18-15db-4e03-b5f2-bdc52d856154	All LAC Queues Telephony EP	NaN	QueueMenu1	NaN	NaN	NaN	2025-03-17 08:14:56	302.0	0 days 00:25:34	False
25	37	25390a18-15db-4e03-b5f2-bdc52d856154	All LAC Queues Telephony EP	NaN	PlayMOH300s	NaN	NaN	NaN	2025-03-17 08:15:17	21.0	0 days 00:25:34	False
26	37	25390a18-15db-4e03-b5f2-bdc52d856154	All LAC Queues Telephony EP	NaN	QueueMenu1	NaN	NaN	NaN	2025-03-17 08:20:19	302.0	0 days 00:25:34	False
27	37	25390a18-15db-4e03-b5f2-bdc52d856154	All LAC Queues Telephony EP	NaN	PlayMOH300s	NaN	NaN	NaN	2025-03-17 08:20:41	22.0	0 days 00:25:34	False
28	37	25390a18-15db-4e03-b5f2-bdc52d856154	All LAC Queues Telephony EP	NaN	NaN	Consumer	NaN	Customer Left	2025-03-17 08:25:25	$\boxed{284.0}$	0 days 00:25:34	False

Calls with an activity in final row

```
termination_row = calls_customer_left.loc[calls_customer_left['Termination Reason'] == 'Customer Left']

termination_row[~termination_row['Activity Name'].isna()]
```

✓ 0.0s

Call ID	Contact Session ID	EP Name	Flow Name	Activity Name	Queue Name	Agent Name	Termination Reason	Activity Start Timestamp	Time Difference	Duration

333 calls were abandoned during hold music

No calls had an activity in their final row

Code for identifying each outcome can be found in appendix

Objective 2

Calls that never enter a queue

117354	8173	9da4f945-76fe-422c-bff5-1acf63c9d723	Legal Housing Menu Telephony EP	NaN	TenantMenu	NaN	NaN	NaN	2025-04-21 10:08:59	42.0	0 days 02:13:29
117386	8173	9da4f945-76fe-422c-bff5-1acf63c9d723	Legal Housing Menu Telephony EP	NaN	HousingMenu	NaN	NaN	NaN	2025-04-21 10:09:35	36.0	0 days 02:13:29
117405	8173	9da4f945-76fe-422c-bff5-1acf63c9d723	Legal Housing Menu Telephony EP	NaN	PreTenantMenu	NaN	NaN	NaN	2025-04-21 10:09:55	20.0	0 days 02:13:29
117426	8173	9da4f945-76fe-422c-bff5-1acf63c9d723	Legal Housing Menu Telephony EP	NaN	TenantMenu	NaN	NaN	NaN	2025-04-21 10:10:37	42.0	0 days 02:13:29
117450	8173	9da4f945-76fe-422c-bff5-1acf63c9d723	Legal Housing Menu Telephony EP	NaN	HousingMenu	NaN	NaN	NaN	2025-04-21 10:11:12	35.0	0 days 02:13:29
117459	8173	9da4f945-76fe-422c-bff5-1acf63c9d723	Legal Housing Menu Telephony EP	NaN	PreTenantMenu	NaN	NaN	NaN	2025-04-21 10:11:32	20.0	0 days 02:13:29
117482	8173	9da4f945-76fe-422c-bff5-1acf63c9d723	Legal Housing Menu Telephony EP	NaN	TenantMenu	NaN	NaN	NaN	2025-04-21 10:12:14	42.0	0 days 02:13:29
117509	8173	9da4f945-76fe-422c-bff5-1acf63c9d723	Legal Housing Menu Telephony EP	NaN	HousingMenu	NaN	NaN	NaN	2025-04-21 10:12:50	36.0	0 days 02:13:29
117518	8173	9da4f945-76fe-422c-bff5-1acf63c9d723	Legal Housing Menu Telephony EP	NaN	PreTenantMenu	NaN	NaN	NaN	2025-04-21 10:13:10	20.0	0 days 02:13:29
117551	8173	9da4f945-76fe-422c-bff5-1acf63c9d723	Legal Housing Menu Telephony EP	NaN	TenantMenu	NaN	NaN	NaN	2025-04-21 10:13:52	42.0	0 days 02:13:29
117572	8173	9da4f945-76fe-422c-bff5-1acf63c9d723	Legal Housing Menu Telephony EP	NaN	HousingMenu	NaN	NaN	NaN	2025-04-21 10:14:27	35.0	0 days 02:13:29
117583	8173	9da4f945-76fe-422c-bff5-1acf63c9d723	Legal Housing Menu Telephony EP	NaN	PreTenantMenu	NaN	NaN	NaN	2025-04-21 10:14:47	20.0	0 days 02:13:29
117601	8173	9da4f945-76fe-422c-bff5-1acf63c9d723	Legal Housing Menu Telephony EP	NaN	TenantMenu	NaN	NaN	NaN	2025-04-21 10:15:29	42.0	0 days 02:13:29
117619	8173	9da4f945-76fe-422c-bff5-1acf63c9d723	Legal Housing Menu Telephony EP	NaN	HousingMenu	NaN	NaN	NaN	2025-04-21 10:16:05	36.0	0 days 02:13:29
117624	8173	9da4f945-76fe-422c-bff5-1acf63c9d723	Legal Housing Menu Telephony EP	NaN	PreTenantMenu	NaN	NaN	NaN	2025-04-21 10:16:24	19.0	0 days 02:13:29
117636	8173	9da4f945-76fe-422c-bff5-1acf63c9d723	Legal Housing Menu Telephony EP	NaN	TenantMenu	NaN	NaN	NaN	2025-04-21 10:17:07	43.0	0 days 02:13:29
117660	8173	9da4f945-76fe-422c-bff5-1acf63c9d723	Legal Housing Menu Telephony EP	NaN	HousingMenu	NaN	NaN	NaN	2025-04-21 10:17:42	35.0	0 days 02:13:29
117670	8173	9da4f945-76fe-422c-bff5-1acf63c9d723	NaN	LegalMenu	NaN	NaN	NaN	NaN	2025-04-21 10:18:09	27.0	0 days 02:13:29
117671	8173	9da4f945-76fe-422c-bff5-1acf63c9d723	Legal Menu Telephony EP	NaN	LegalMenu2	NaN	NaN	NaN	2025-04-21 10:18:09	0.0	0 days 02:13:29
117672	8173	9da4f945-76fe-422c-bff5-1acf63c9d723	Legal Menu Telephony EP	NaN	NaN	NaN	NaN	Customer Left	2025-04-21 10:18:09	0.0	0 days 02:13:29

This example reflects when the caller is unable to find the correct menu and queue for their legal issue as this caller spent roughly two hours alternating between the Housing, Tenant, and Pre-Tenant menus.

45,396 abandoned calls never entered a queue

Code for identifying each outcome can be found in appendix

Objective 2

2025/03/24	08:34:21 AM	Termination Reason	Agent Left
2025/03/24	08:36:21 AM	Queue Name	Family

Issues we Faced + Next Steps

Issues:

- Unsure of what criteria to use to draw distinction between different cases where caller never enters queue
- Uncertainty of meaning of certain termination reasons
    - ‘RONA_TIMER_EXPIRED’
    - ‘RONA Timer Expired’
    - ‘CONTACT_CALLBACK_IN_PROGRESS’
- Some cases of activity after a termination reason
    - The agent left but two minutes later the call is still in the Family Queue

Next Steps:

- Refine filtering of calls that never enter a queue to exclude very brief calls (e.g. customer changes their mind about calling) which should be a neutral outcome
- Add outcome and/or termination reason as a column to help with eventual visualization
- Analyzing the proportion of calls that receive a successful callback, and how many calls it took to reach this successful outcome

Objective 3

Nikole

Question: Does the Pre-Tenant Menu actually reduce caller load?

The Pre-Tenant Menu informs callers about eviction summons procedures.

Objective: Identify how many callers abandon after the Pre-Tenant Menu vs continue to the Tenant Menu.

Dataset used: CAR only.

Methodology:

- Identified all sessions that reached PreTenantMenu and TenantMenu by applying filter in the variable ‘Activity Name’ in CAR.
- Remove duplicates
- Tracking Call Paths

Objective 3

Nikole

Data Issues/Cleaning

- Some callers looped through the Pre-Tenant Menu multiple times.
- Decided to kept only one record per ‘Contact Session ID’ to ensure each caller was counted once.
- This step avoids overcounting callers who repeatedly re-entered the same menu.

`Contact Session ID`	`EP Name`	`Flow Name`	`Activity Name`	Activity Start Times…¹	`Queue Name`	`Agent Name`	`Termination Reason`
<chr>	<chr>	<chr>	<chr>	<dttm>	<chr>	<chr>	<chr>
0042f5e6-6c86-4bc1-84a5-4bf…	Legal Ho…	N/A	PreTenantMenu	2025-01-14 13:19:29	N/A	N/A	NA
0268aa9d-4961-460b-a996-423…	Legal Ho…	N/A	PreTenantMenu	2025-01-15 13:42:00	N/A	N/A	NA
0317b528-60f8-4abe-a7b1-787…	Legal Ho…	N/A	PreTenantMenu	2025-01-15 09:35:34	N/A	N/A	NA
0317b528-60f8-4abe-a7b1-787…	Legal Ho…	N/A	PreTenantMenu	2025-01-15 09:37:12	N/A	N/A	NA
0317b528-60f8-4abe-a7b1-787…	Legal Ho…	N/A	PreTenantMenu	2025-01-15 09:39:45	N/A	N/A	NA
041f1006-e1f1-4d7e-afe7-602…	Legal Ho…	N/A	PreTenantMenu	2025-01-16 09:54:04	N/A	N/A	NA
0522c06b-b5b2-4fc0-8b22-733…	Legal Ho…	N/A	PreTenantMenu	2025-01-14 10:44:49	N/A	N/A	NA
07b2d48b-7d9c-4e84-a41f-35d…	Legal Ho…	N/A	PreTenantMenu	2025-01-16 15:18:01	N/A	N/A	NA
082d8789-354a-4196-ba2b-258…	Legal Ho…	N/A	PreTenantMenu	2025-01-14 10:57:13	N/A	N/A	NA
082d8789-354a-4196-ba2b-258…	Legal Ho…	N/A	PreTenantMenu	2025-01-14 10:58:52	N/A	N/A	NA

Objective 3

Determining calls paths

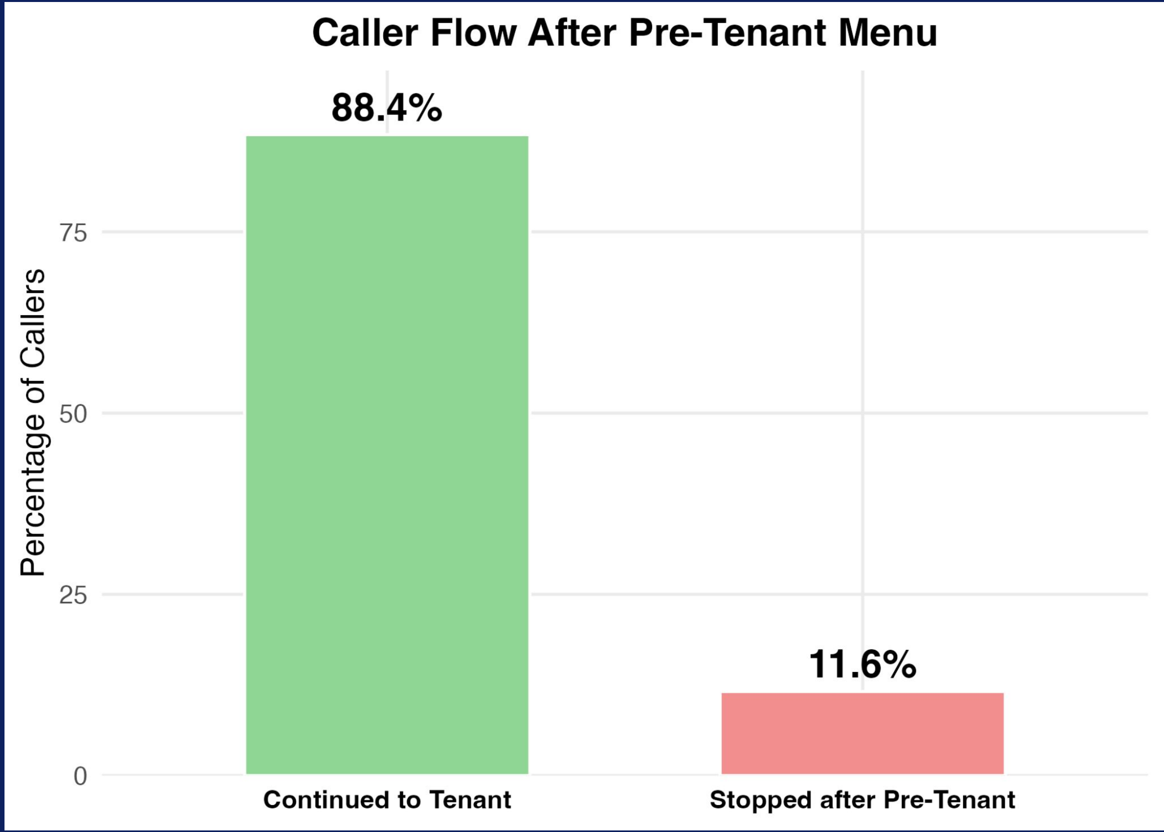
- Compared Pre-Tenant Menu calls against Tenant Menu calls.
- Tagged each Pre-Tenant session as:
  - Continued to Tenant (caller navigated onward)
  - Stopped after Pre-Tenant (caller ended there)

```
pre_tenant_calls_categorized <- pre_tenant_calls %>%
  mutate(path_status = if_else(contact_session_id %in% tenant_calls$contact_session_id,
                               "continued_to_tenant",
                               "stopped_after_pretenant"))
```

Objective 3

Main Takeaways

- PreTenant Menu is largely ineffective at doing its filtering purpose
- Low self termination/abandon of calls
- PreTenant Menu fails its filtering purpose
  - Almost 9 out of 10 callers continue to TenantMenu



Objective 3

Nikole

Recommendations:

- Consider removing or shortening the Pre-Tenant Menu.
- Alternatively, move it after the main Tenant Menu for better triage.
- Monitor abandonment again post-change to evaluate impact.

Thank you!

Appendix

Objective 1

Call Data Integration and Preparation Workflow

Combined the data into one large dataset →

```r
combined_data <- map_dfr(file_list, read_and_tag)

combined_data <- combined_data %>%
  mutate(
    activity_datetime = ymd_hms(`Activity Start Timestamp`),
    weekday_number = wday(activity_datetime),
    Weekend_Weekday = ifelse(weekday_number %in% c(1, 7), "Weekend", "Weekday")
  )
```

Adjust timestamps to Chicago timezone →

```r
all_calls |> janitor::clean_names() |>
  mutate(report_time = with_tz(report_time, tzone = 'America/Chicago'),
         month = month(report_time, label = TRUE, abbr = FALSE),
         year = year(report_time),
         month_year = paste0(month, ' ', year),
         )
```

Extract only valid inbound calls from All Calls — calls lasting more than 0 minutes, with direction "TERMINATING", and PSTN vendor "CallTower" →

```r
all_calls |>
  filter(duration > 0,
         direction == "TERMINATING",
         pstn_vendor_name == 'CallTower') |>
  distinct(correlation_id, .keep_all = TRUE) |>
```

Count distinct "contact_session_id" in All Calls and "correlation_id" in CAR →

```r
car |>
  distinct(contact_session_id, .keep_all = TRUE) |>
```

Objective 2

Code demonstrating queue timeout analysis + number of courtesy callbacks

```
# filter by desired end condition
queue_time = df_long[df_long['Activity'] == 'Queue Timeout']

# get the full journey for those calls
filtered_journeys = df_long[df_long['Contact Session ID'].isin(queue_time['Contact Session ID'])]

filtered_journeys.groupby("Contact Session ID").tail()
```

```
# example of how to manually get the number of courtesy callbacks
filtered_journeys[(filtered_journeys['Contact Session ID'] == "0d5dec0a-ec28-4bb0-a728-5f848b543d8d")
                  & (filtered_journeys['Activity'] == "Courtesy Callback Telephony EP")]
```

✓ 0.0s

Objective 2

Code for analyzing System Disconnect

```
last_steps = df_long.groupby('Contact Session ID').tail(1)

# filter by desired end condition
system_dis = last_steps[last_steps['Activity'] == 'System disconnected the contact']

# get the full journey for those calls
filtered_journeys = df_long[df_long['Contact Session ID'].isin(system_dis['Contact Session ID'])]

filtered_journeys.groupby("Contact Session ID").nth(2)
```

✓ 0.0s

Objective 2

Code for Identifying Calls Abandoned during Hold Music

```python
# Finding Contact Session IDs where caller listened to hold music
hold_music_calls = calls_customer_left.loc[calls_customer_left['Activity Name'] == 'PlayMOH300s', 'Contact Session ID']

# Filtering to only include those Contact Session IDs
music_calls = calls_customer_left.loc[calls_customer_left['Contact Session ID'].isin(hold_music_calls), :]
```

[11] ✓ 0.0s

```python
# Sort by Call ID so all steps in a customer's call journey will be grouped together (i.e. no non-consecutive entries) & follow progression of their steps based on time
music_calls.sort_values(by=['Call ID', 'Activity Start Timestamp'], inplace=True)

# Resetting index so that indices will correspond to their new order
music_calls.reset_index(drop=True, inplace=True)
```

[12] ✓ 0.0s

```python
# Function to check if the user left while listening to the hold music
def calls_abandoned_MOH(music_calls, term_col='Termination Reason', term_val='Customer Left',
                        activity_col='Activity Name', activity_val='PlayMOH300s',
                        time_col='Time Difference', time_threshold=300):
    return ((music_calls[term_col] == term_val)
            & (music_calls[activity_col].shift(1) == activity_val)
            & (music_calls[time_col] < time_threshold)
           ).sum()

# Applying function to calls
abandoned_during_MOH = calls_abandoned_MOH(music_calls)

#
print('Number of Calls Abandoned while Listening to Hold Music:', int(abandoned_during_MOH))

# Printing total number of calls that listened to hold music for comparison
print('Total Number of Abandoned Calls that Listened to Hold Music:', music_calls['Call ID'].nunique())
```

[76] ✓ 0.0s

```
Number of Calls Abandoned while Listening to Hold Music: 343
Total Number of Abandoned Calls that Listened to Hold Music: 3333
```

Objective 2

Code for Identifying Calls that Never Entered a Queue

```python
# Grouping by Call ID to check that Queue Name is NaN for all rows of call -> never enters a queue
no_queue_calls = calls_customer_left.groupby('Call ID')['Queue Name'].apply(lambda x: x.isna().all())
no_queue_call_ids = no_queue_calls[no_queue_calls].index

# Filtering dataframe to only include Call IDs of calls that don't enter queue
never_enter_queue = calls_customer_left[calls_customer_left['Call ID'].isin(no_queue_call_ids)]

print('Number of Abandoned Calls that Never Enter a Queue:', never_enter_queue['Call ID'].nunique())
```

[75] ✓ 1.4s

```
Number of Abandoned Calls that Never Enter a Queue: 45396
```

Objective 3

Code for Identifying calls that reached PreTenant Menu and Tenant Menu →

```r
# All sessions that reached PreTenantMenu
pre_tenant_calls <- car %>%
  filter(activity_name == "PreTenantMenu") %>%
  distinct(contact_session_id, .keep_all = TRUE)

# All sessions that reached TenantMenu
tenant_calls <- car %>%
  filter(activity_name == "TenantMenu") %>%
  distinct(contact_session_id, .keep_all = TRUE)
```

Categorizing calls from the Pretenant session that later reached the Tenant Menu →

```r
# Tag each PreTenant session by whether it later reached TenantMenu
pre_tenant_calls_categorized <- pre_tenant_calls %>%
  mutate(path_status = if_else(contact_session_id %in% tenant_calls$contact_session_id,
                               "continued_to_tenant",
                               "stopped_after_pretenant"))

# Quick summary
pre_tenant_summary <- pre_tenant_calls_categorized %>%
  count(path_status) %>%
  mutate(percent = round(100 * n / sum(n), 1))

print(pre_tenant_summary)
```